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Quantitative systems toxicology approach that integrates PBPK and core hepatic metabolism: a case study with valproic acid

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New approach methodologies (NAMs) are advancing the reduction of animal testing by promoting human-relevant safety assessments across diverse applications, including cosmetics, pharmaceuticals, and environmental chemicals. Among these methodologies, computational models like quantitative systems toxicology (QST) have emerged as powerful tools, enabling the simulation of mechanisms underlying drug (or chemical) induced toxicity to predict potential adverse outcomes. Valproic acid (VPA), a treatment for epilepsy, convulsions and bipolar disorder, is associated with a risk of drug-induced liver injury. The mechanism of hepatotoxicity is not fully understood, although VPA's competitive inhibition of fatty acid metabolism via carnitine palmitoyl transferase 1 (CPT1) in the liver is a leading hypothesis. In this study, we employed a QST approach by integrating a physiologically-based pharmacokinetic model of VPA with the large-scale liver metabolism model HEPATOKIN1 to evaluate whether simulated VPA dosing with increasing CPT1 inhibition predicts clinically relevant markers of lipid metabolism disruption and hepatic triacylglyceride (TAG) accumulation, an early marker for hepatic steatosis. The integrated model predicted a dose-dependent increase in hepatic TAG as a result of the competitive inhibition of CPT1 by VPA, qualitatively consistent with clinical observations. Notably, explicit inclusion of PPAR α -mediated regulatory effects on key liver enzymes was essential to ensure biologically consistent outcomes in liver metabolism. These results highlight the necessity of including relevant regulatory pathways in QST applications to achieve credible and physiologically relevant safety predictions.

KEYWORDS

hepatic steatosis, new approach methodologies, PPAR α , quantitative systems toxicology, valproic acid

1 Introduction

Valproic acid (VPA) is a wide-ranging therapeutic agent primarily utilized in the treatment of epilepsy and convulsions, but also used to treat migraine, schizoaffective disorder, and bipolar disorder for its mood-stabilising properties (Tomson et al., 2016; Lagace et al., 2004). A significant clinical concern associated with VPA is its potential to cause liver damage; however, the exact mechanisms underlying VPA-induced hepatotoxicity remains poorly understood (Ezhilarasan and Mani, 2022; Chang and Abbott, 2006). Among

the various hypotheses, one prominent theory suggests that, as a branched, short-chain fatty acid, VPA competitively inhibits catabolism of fatty-acids through the β -oxidation pathway, an essential process in fatty acid metabolism (Silva et al., 2008). Reduced fatty-acid catabolism through this pathway can result in metabolic-dysfunction and drug-induced liver injury (DILI). Clinically, this can present as alterations in metabolic markers - plasma triglycerides (TAG), lipoproteins, and total serum cholesterol - and associated hepatic steatosis, potentially progressing to elevated liver injury markers such as aspartate aminotransferase (AST), alanine aminotransferase (ALT), and bilirubin (Nanau and Neuman, 2013; Ahanger et al., 2017).

The quantitative systems toxicology (QST) approach aims to quantitatively understand the adverse effects of chemicals on a living organism, from molecular alterations to phenotypical observations, through integrating computational modelling and simulation, and experimental methods (Beattie et al., 2024; Goldring et al., 2025). More established approaches under the systems biology umbrella, such as quantitative systems pharmacology (QSP) and physiologically-based pharmacokinetic (PBPK) modelling apply the same computational tools to mechanistically understand pharmacodynamics and pharmacokinetics, respectively (Jones and Rowland-Yeo, 2013). The evermore routine application of these approaches to inform decision making in the drug development pipeline and their increasing acceptance by regulatory authorities (Tan et al., 2018; El-Khateeb et al., 2021) is reflective of an industry wide adoption of model-informed drug discovery and development (MID3).

Several mathematical models of varying scale and mechanistic resolution have been developed to simulate liver function and chemical-induced liver injury. Notable examples include DILIsym[®] and HEPATOKIN1 (Berndt et al., 2018) among numerous others (Siler, 2022; Schleicher and Dahmen, 2018; Dutta-Moscato et al., 2014; Ashworth et al., 2016). DILIsym[®] is a multi-scale mechanistic model that integrates cellular toxicity with systemic liver injury biomarkers such as ALT, AST, and bilirubin, and has been validated against exemplar compounds known to cause DILI (Eichenbaum et al., 2020), including VPA-induced liver injury via mitochondrial dysfunction, bile acid transport inhibition, and production of reactive oxygen species; however, it does not specifically model hepatic steatosis (Lakhani et al., 2023). Expanding on this framework, NAFLDsym[®] (Siler, 2022) incorporates fatty acid dynamics, inflammatory cells, and hepatic stellate function to simulate metabolic dysfunction-associated fatty liver disease (MAFLD, formerly non-alcoholic fatty liver disease) and metabolic dysfunction-associated steatohepatitis (MASH, formerly non-alcoholic steatohepatitis) (Kanwal et al., 2024). This expansion has enabled the prediction of treatment efficacy for both MAFLD and MASH. In contrast, the HEPATOKIN1 (Berndt et al., 2018) model focuses on modelling hepatocyte functions at the molecular and cellular scale, enabling simulation of cell-level processes that contribute to DILI. HepatoNet1 (Gille et al., 2010), originally developed as a metabolic network model, was subsequently expanded into HEPATOKIN1, resulting in a highly detailed kinetic model of hepatocyte metabolism at the genome scale, comprehensively representing metabolic pathways for glycogen, fructose, fatty acids, ketones, and other substrates. This genome-scale resolution enables the incorporation of specific

cellular-level mechanisms of DILI in relation to these metabolic pathways and facilitates the simultaneous tracking of numerous metabolic substrates, offering insights not present in the multi-scale models DILIsym[®] or NAFLDsym[®].

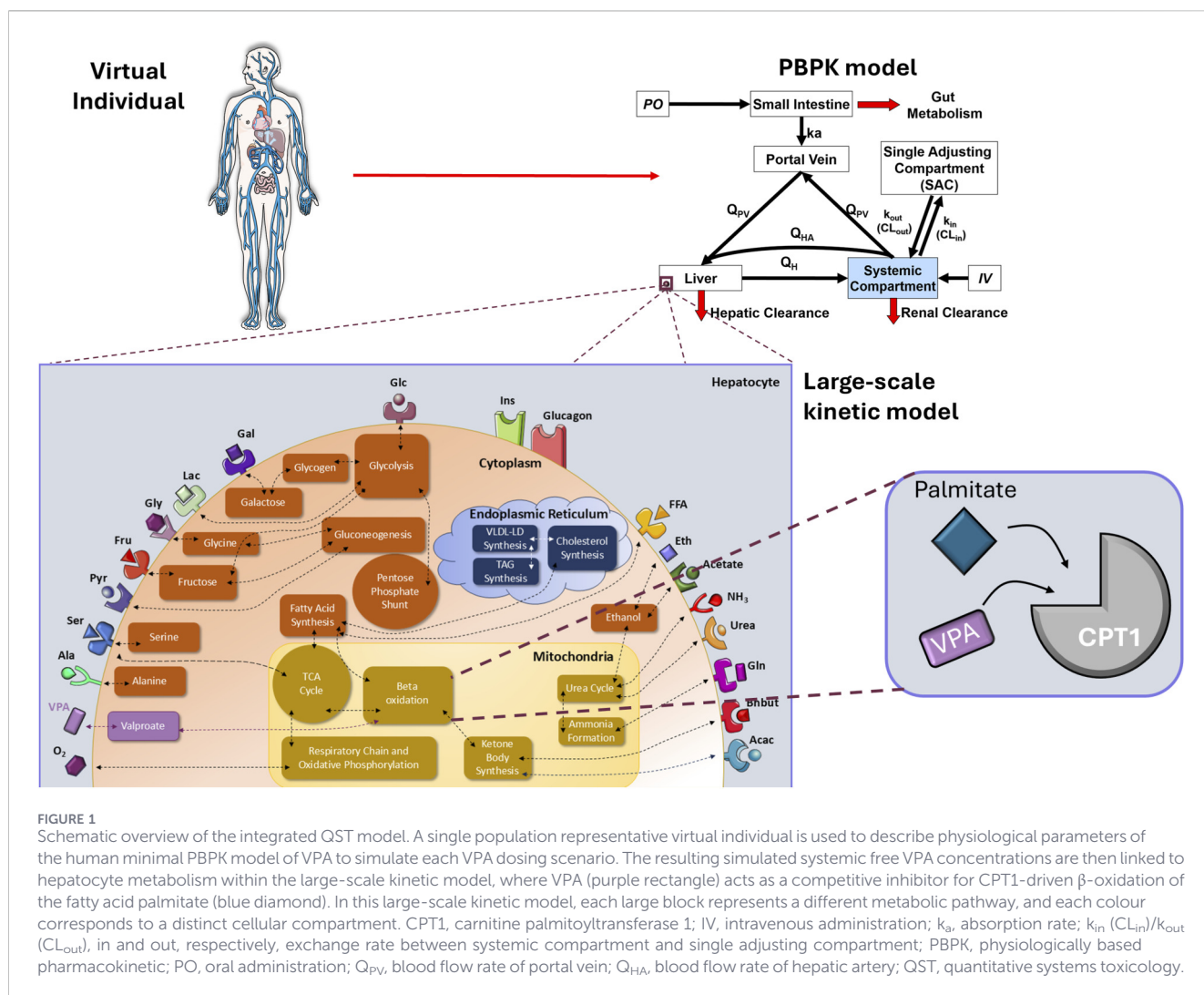
It is vital to recognize model limitations to ensure appropriate use in DILI assessment. A notable limitation among the models discussed, including HEPATOKIN1 (Berndt et al., 2018), is the absence of relevant adaptive feedback mechanisms, which are fundamental to the non-linear behaviour of biological systems and the regulation of processes such as inflammation (Qian et al., 2015), differentiation (Laudadio et al., 2012), and metabolism (Barbier et al., 2003; Kersten and Stienstra, 2017). The transcription factor peroxisome proliferator-activated receptor α (PPAR α) plays a vital role in lipid metabolism by regulating the expression of several enzymes essential for hepatocyte metabolism. PPAR α is particularly integral in modulating the β -oxidation of fatty acids, acting as a critical regulatory node in lipid homeostasis (Dreyer et al., 1993; Minnich et al., 2001). Its influence extends to the regulation of gluconeogenesis and amino acid metabolism and may be especially important when simulating the effects of chemical-induced disruption of hepatic fatty acid metabolism (Todisco et al., 2022; Kersten, 2014; Gutgesell et al., 2009; Li et al., 2014; Maldonado et al., 2018). Previous systems modelling work used a quasi-steady state approach to simulate dynamic regulation of hepatic metabolism; using the HepatoNet1 (Gille et al., 2010) model, showed that increased flux towards intracellular TAG in response to fatty-acid treatment was dependent on PPAR α -mediated regulation (Maldonado et al., 2018).

In this study, we develop a multi-scale QST model integrating the cellular-level liver metabolic model, HEPATOKIN1, with a human PBPK model of VPA to simulate VPA-induced disruption of the β -oxidation pathway and predict hepatotoxic potential. The presence of non-linear plasma concentrations of VPA with increasing dose, driven by concentration-dependent plasma protein binding (FDA, 2016; Zaccara et al., 1988), underscores the importance of PBPK to accurately predict VPA concentrations in the liver. This approach includes a semi-quantitative evaluation of PPAR α 's regulatory effects, serving as a case study to demonstrate the role of regulatory feedback mechanisms as an essential consideration for application of QST models.

2 Methods

2.1 Minimal physiologically based pharmacokinetic model for VPA

We developed a minimal PBPK (mPBPK) model using the Simcyp Simulator (v23; Certara Inc., Simcyp Division Sheffield, UK) to predict the free plasma concentration of VPA in humans, as previously published in an Organisation for Economic Co-operation and Development (OECD) case study (Escher et al., 2020), with parameter values detailed in [Supplementary Table S1](#). PBPK models simulate drug absorption, distribution, metabolism, and elimination (ADME) using physiological compartments representing organs like blood, liver, and intestines, calibrated by physiological parameters such as body weight, liver weight, and blood volume. The mPBPK approach uses minimal



compartmentation to adequately model drug pharmacokinetics (Jones and Rowland-Yeo, 2013) (Figure 1), typically including gut absorption, liver metabolism, and renal clearance. The details of the PBPK simulator and Simcyp's quality assurance system are provided elsewhere (Ke et al., 2016; Rowland Yeo et al., 2024; Ezurike et al., 2022). The VPA PBPK model was parameterised using an in vitro-to-in vivo extrapolation (IVIVE) approach, and its predictive performance was verified against published clinical data (Georgoff et al., 2018). VPA exhibits concentration-dependent protein binding (FDA, 2016), which is included into the mPBPK model by accounting for its binding to plasma proteins, specially human serum albumin (HSA), using the equation described previously (Berezhkovskiy, 2007):

$$f_u(t) = \frac{1 - \frac{\gamma_{PT}}{BP} - \frac{K_d}{BP} + \sqrt{\left(1 + \frac{\gamma_{PT}}{BP} + \frac{K_d}{BP}\right)^2 - 4 \frac{\gamma_{PT}}{BP}}}{2} \quad (1)$$

where $f_u(t)$ denotes concentration-dependent fraction of unbound VPA in plasma, γ_{PT} denotes total albumin concentration (assumed constant), K_d denotes the drug binding affinity to HSA, $val_{tot}(t)$

denotes the total VPA concentration in blood, and BP denotes the blood to plasma ratio of VPA.

To verify the model against clinical data, we simulated a single intravenous dose of VPA infused over 1 h with 30 mg/kg or 130 mg/kg across 72 h in a population of healthy volunteers (Sim-Healthy, $N = 8$ for 10 trials) with 8% females to mirror the dosing and population of the clinical study (Georgoff et al., 2018). The Simcyp Simulator generates unique virtual individuals by sampling from a range of physiological parameters as described previously (Curry et al., 2024). The physiological parameters that vary between virtual individuals and are relevant to the mPBPK model for VPA are listed in Supplementary Table S3. In subsequent simulations, we predicted the systemic VPA concentration in a single population representative virtual individual of a 'healthy' individual following once-daily oral doses of 100, 200, 500, 1,000, 2,000, or 5,000 mg.

2.2 HEPATOKIN1 hepatocyte metabolism model reconstruction

We implemented a kinetic model of core hepatocyte metabolism, HEPATOKIN1 (Berndt et al., 2018), in Simcyp

Designer software (V3; Certara Predictive Technologies, Sheffield, UK) (Matthews et al., 2023). Initially, the model implementation was verified to reproduce outcomes identical to those of the published model provisioned in SBML format by the authors (Supplementary Figure S1A), in line with previously recommended reproducibility assessments (Tiwari et al., 2021). Furthermore, the reconstructed model was evaluated for its ability to predict cytosolic glycogen concentration during the transition from the fasted (glucose = 4 mM) to the fed (glucose = 8 mM) state (Supplementary Figure S1B), as well as diurnal profiles of glutamate (Supplementary Figure S1C) and lactate (Supplementary Figure S1D) against experimental data, recovering the performance demonstrated by the published model (Berndt et al., 2018).

2.3 Integration of VPA mPBPK and HEPATOKIN1

Simcyp Designer software was used to integrate the mPBPK model for VPA, developed in the Simcyp Simulator, with the reconstructed large-scale model HEPATOKIN1 (hereafter referred to as the “integrated model”). The models were linked by predicted VPA systemic concentration, as illustrated in Figure 1. VPA is considered an exemplar Extended Clearance Classification System (ECCS) class 1A compound with high permeability, ~90% contribution of metabolic hepatic clearance to systemic clearance, and no significant contribution of hepatic uptake or efflux transporters (Varma et al., 2015): hence, we assumed rapid equilibrium of unbound concentration between plasma and hepatic intracellular water. From the mPBPK predicted VPA systemic concentration, intracellular free VPA concentration (val_{cyt}) within hepatocytes was estimated by considering factors such as plasma protein binding ($f_u(t)$), BP, and diffusion from plasma to the hepatocyte intracellular space (Supplementary Table S1). The equation to determine val_{cyt} is described below:

$$\frac{d[val_{cyt}(t)]}{dt} = D \left(\frac{[val_{tot}(t)] f_u(t)}{BP} - val_{cyt}(t) \right) \quad (2)$$

where val_{tot} denotes the total VPA concentration in blood, D denotes the diffusion rate between blood and hepatocytes of 3.6 h^{-1} (Berndt et al., 2018), and $f_u(t)$ is calculated using Equation 1.

The resulting val_{cyt} from Equation 2 then acts as a substrate to relevant metabolic enzymes within the metabolic model for metabolism and elimination. val_{cyt} competitively inhibited the enzyme carnitine palmitoyltransferase I (CPT1) in accordance with previously established mechanisms (Berndt et al., 2018) and as reported previously in literature (Ezhilarasan and Mani, 2022; Aires et al., 2010) (Figure 1). In the model, CPT1 plays a vital role in the β -oxidation pathway as it is responsible for the elimination of endogenous fatty acids such as palmitate. The conversion of val_{cyt} to valproyl coenzyme A ($valCoA_{cyt}$) by valproylcoa synthetase (VCS), followed by competitive inhibition of CPT1, is described in the equations below as represented in HEPATOKIN1. For all remaining equations and parameter definitions not explicitly shown here, please refer to the original HEPATOKIN1 model (Berndt et al., 2018).

$$\frac{d[valCoA_{cyt}(t)]}{dt} = V_{max}^{VCS} \frac{[val_{cyt}(t)]}{[val_{cyt}(t)] + K_m^{VCSval_{cyt}}} \frac{[ATP_{cyt}(t)]}{[ATP_{cyt}(t)] + K_m^{VCSATP_{cyt}}} \frac{[CoA_{cyt}(t)]}{[CoA_{cyt}(t)] + K_m^{VCSCoA_{cyt}}} \quad (3)$$

Equation 3 describes conversion of val_{cyt} to $valCoA_{cyt}$ where, V_{max}^{VCS} is the maximum velocity of VCS ($3.6 \text{ mol/m}^3\text{h}$). $K_m^{VCSval_{cyt}}$ is the Michaelis constant for val_{cyt} with VCS (1.65 mol/m^3); $[ATP_{cyt}(t)]$ is the cytosolic ATP concentration, and the corresponding $K_m^{VCSATP_{cyt}}$ for ATP with VCS denotes its Michaelis constant (0.6 mol/m^3); $[CoA_{cyt}(t)]$ is the cytosolic coenzyme A concentration, and the corresponding $K_m^{VCSCoA_{cyt}}$ for coenzyme A with VCS (0.0025 mol/m^3) denotes its Michaelis constant.

$$v_{CPT1} = V_{max}^{CPT1} \frac{[c16CoA_{cyt}(t)]}{[c16CoA_{cyt}(t)] + K_m^{c16CoA_{cyt}}} \frac{[carnitine_{cyt}(t)]}{[carnitine_{cyt}(t)] + K_m^{carnitine_{cyt}}} \quad (4)$$

Equation 4 describes v_{CPT1} , which is the flux of CPT1 ($\text{mol/m}^3\text{h}$) enzyme driven removal rate of cytosolic palmitate coenzyme A [$c16CoA_{cyt}(t)$]. V_{max}^{CPT1} is the maximum velocity of CPT1 ($1.8 \text{ mol/m}^3\text{h}$). Cytosolic carnitine, $[carnitine_{cyt}(t)]$, is the CPT1 substrate concentration, and $K_m^{carnitine_{cyt}}$ is the Michaelis constant for carnitine with CPT1 (0.032 mol/m^3). $K_m^{c16CoA_{cyt}}$ is the Michaelis constant for [$c16CoA_{cyt}(t)$] with CPT1, calculated using Equation 5 below to describe VPA-mediated inhibition.

$$K_m^{c16CoA_{cyt}} = K_0^{c16CoA_{cyt}} \left(1 + \frac{[valCoA_{cyt}(t)]}{K_i^{valCoA_{cyt}}} \right) \times \left(1 + \frac{[malCoA2_{imm}(t)]}{K_i^{malCoA2_{imm}}} \right) \quad (5)$$

$K_0^{c16CoA_{cyt}}$ is the basal Michaelis constant for [$c16CoA_{cyt}(t)$] binding with CPT1 (0.03 mol/m^3), which is scaled by competitive inhibition from [$valCoA_{cyt}(t)$] and inhibitory constant $K_i^{valCoA_{cyt}}$ (0.057 mol/m^3) and malonyl coenzyme A (malCoA2) [$malCoA2_{imm}(t)$] and inhibitory constant $K_i^{malCoA2_{imm}}$ (0.0025 mol/m^3).

The integrated model was simulated with repeated oral doses of VPA, ranging from 100 to 5,000 mg, at specific dose levels of 100, 200, 500, 1,000, 2,000, and 5,000 mg. Doses of 100 mg, 1,000 mg, and 5,000 mg were selected as representative scenarios for low, medium, and high dosing levels. This dose selection was based on an initial recommended dose range of 10–15 mg/kg/day (FDA, 2016) and the absence of safety guidelines for valproate doses exceeding 60 mg/kg/day (FDA, 2016). The integrated model was simulated over a 5-day (120 h) period, beginning with an initial 24-h phase without VPA dosing to allow core metabolites in HEPATOKIN1 to stabilize in a steady state. This was followed by a regimen of once-daily oral VPA administration for the remaining 4 days (96 h).

2.4 PPAR α -mediated regulation

We identified metabolic reactions within the metabolic component of the integrated model that are reported in the literature to be regulated by the transcription factor PPAR α

(Supplementary Table S2). The flux of these metabolic reactions was scaled by estimating the levels of ‘activated’ PPAR α using Equation 6 below. Here, we assume that PPAR α activation is solely driven by cytosolic palmitate ($c16_{cyt}$). Although oleic acid is recognised as the most abundant fatty acid in healthy human liver, palmitic acid is most prominently associated with adverse metabolic outcomes, including dyslipidaemia, and hyperglycaemia (Kotronen et al., 2010; Carta et al., 2017). This simplified assumption, consistent with HEPATOKIN1, was adopted to manage the complexity of modelling fatty acid species and to exemplify VPA-induced hepatotoxicity. It was modelled using a Hill function, as described: the maximum production rate, $\Phi_{max} = 0.5\ h^{-1}$ (assuming average human protein production rate (Shamir et al., 2016)); the half-maximal $c16_{cyt}$ concentration, $\Phi_{50} = 5 \times 10^{-5}\ mM$ (arbitrarily chosen), Hill’s coefficient, $n = 0.2$ (arbitrarily chosen), and the PPAR α degradation rate $\delta_{PPAR\alpha} = 0.1008\ h^{-1}$ (assuming average human protein degradation rate (Eden et al., 2011)).

$$\frac{dPPAR\alpha}{dt} = \frac{\Phi_{max}c16_{cyt}^n}{\Phi_{50}^n + c16_{cyt}^n} - \delta_{PPAR\alpha}PPAR\alpha \quad (6)$$

The values for Φ_{50} and Hill’s coefficient were arbitrarily chosen due to the absence of experimental data to inform a data-driven value. Hence, we performed a sensitivity analysis on both parameters within reasonable ranges informed by diurnal concentrations from HEPATOKIN1 to assess the impact of varying these parameters on the PPAR α profile at baseline values in the absence of VPA dosing (Supplementary Figures S2, S3). Varying Φ_{50} between 1×10^{-4} and $1 \times 10^{-6}\ mM$ concentrations (basal $c16_{cyt}$ concentrations fluctuate between 1.57×10^{-5} and $5.56 \times 10^{-6}\ mM$) invariably led to an increase in PPAR α values above 1, regardless of the specific value selected. However, when we examined Hill’s coefficient values between 0 and 0.8, not all values for Hill’s coefficient resulted in PPAR α values exceeding 1; for example, at $n = 0.8$, parts of the PPAR α profile fell below 1, implying a reversal in the direction of the PPAR α ’s scaling effect on enzymes. Since our assumption is that PPAR α exerts a linear scaling effect on enzyme kinetics, we chose a Hill’s coefficient of 0.2 that consistently maintained PPAR α values above one to ensure the directionality of upregulation and downregulation on affected enzymes. Although variations in these values may result in quantitatively distinct PPAR α levels, the directional change in target regulation and the ensuing qualitative effects are expected to remain unchanged.

3 Results

3.1 Impact of VPA dosing on lipid metabolism without PPAR α -mediated regulation

To ensure human relevance of the integrated model, we devised a human PBPK framework to model VPA pharmacokinetics, which was verified against observed human data. This human relevance stems partly from the incorporation of physiology-based parameters sourced from literature, including the blood to plasma ratio and human serum albumin binding affinity (see Methods). Consistent with its human relevance, the VPA PBPK model quantitatively

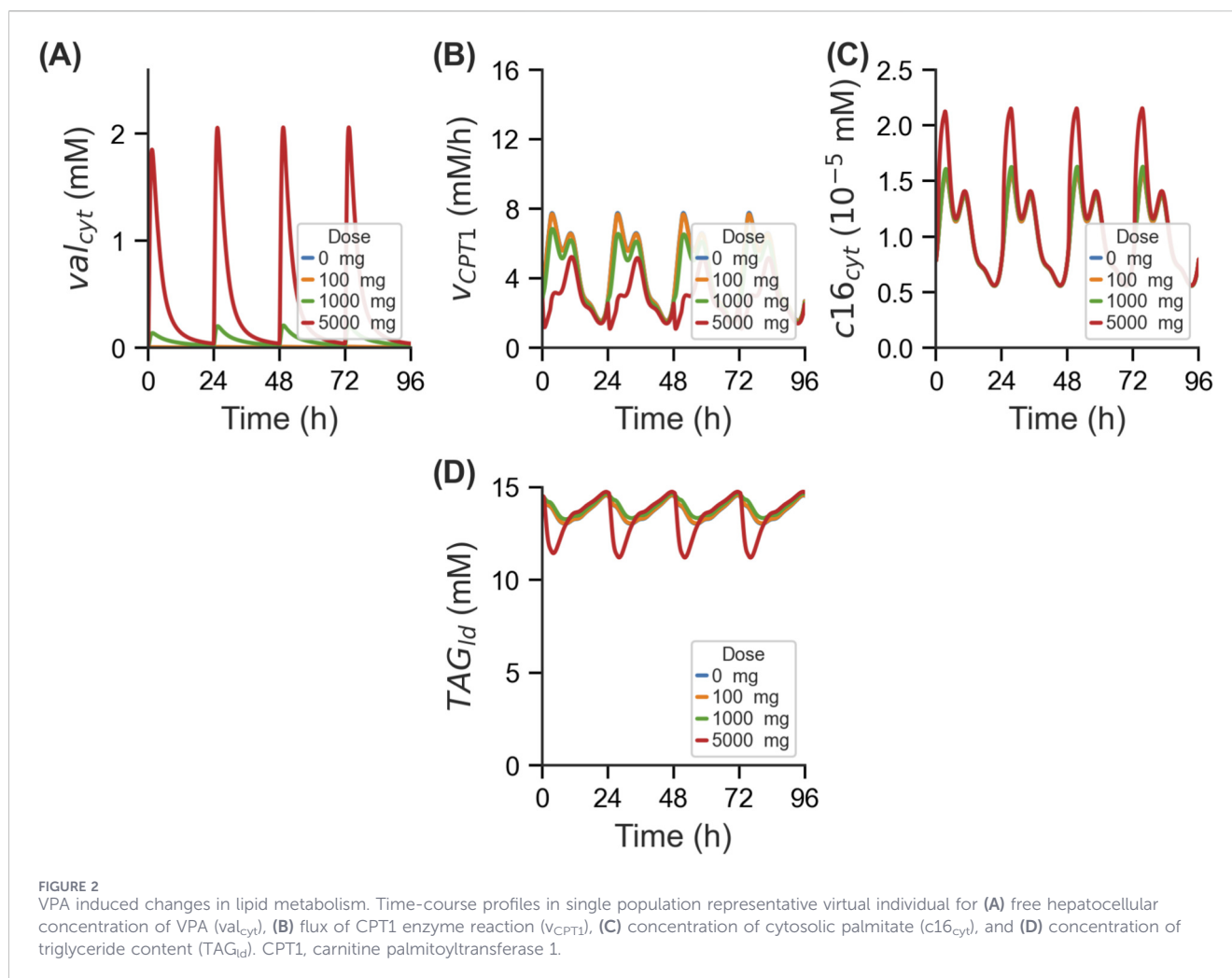
replicates clinically observed plasma concentrations of VPA following intravenous doses of 30 and 130 mg/kg administered over 1 h (Supplementary Figure S4). For instance, the model’s predicted mean and 95% confidence interval, which incorporates population variability in physiology, recovers the measured observations well, such as peak concentration (C_{max}) and subsequent elimination from plasma. Following this validation, the model’s predicted systemic VPA concentration was used to calculate the free VPA concentration in hepatocytes. This effective concentration was then applied to stimulate the HEPATOKIN1 model within the integrated model, thereby enhancing its relevance to human liver tissue.

This integrated model was employed to investigate the effects of increasing doses of VPA on lipid metabolism. We simulated the model with oral VPA administration once every day over a 72-h period in a single population representative virtual individual, as detailed in the Methods section. As expected, intracellular concentration of VPA within hepatocytes (val_{cyt}) levels rose with higher VPA doses (Figure 2A). Notably, carnitine palmitoyltransferase 1 (CPT1) flux, v_{CPT1} , decreased with increased VPA dosing due to the competitive inhibition by VPA (Figure 2B), resulting in elevated concentrations of cytosolic palmitate, $c16_{cyt}$ (Figure 2C). Since palmitate is a cofactor of TAG, including those found in lipid droplets (TAG_{ld}) (Figure 2D), increasing VPA dosing generally led to elevated TAG_{ld} levels. Surprisingly, at the highest evaluated dose of 5,000 mg, a reduction in TAG_{ld} was predicted, suggesting a possible threshold effect or alternative metabolic regulation at extremely high VPA concentrations occurring from the integrated model.

To elucidate the mechanism responsible for the reduction in TAG_{ld} predicted at higher doses, we investigated additional TAG cofactors such as cytosolic long chain acyl coenzyme A ($c16CoA_{cyt}$) (Supplementary Figure S5A) and cytosolic coenzyme A (CoA_{cyt}) (Supplementary Figure S5B). In simulations at the highest dose of 5,000 mg, CoA_{cyt} reduced to near zero concentrations, indicating complete depletion of the cofactor. This is likely contributing to the reduction of TAG_{ld} below baseline levels predicted only at high doses as the cofactor is completely depleted to make TAG. However, it is physiologically improbable for metabolites to be consistently depleted, as feedback mechanisms would typically prevent this; indeed, elevated TAG levels are reported to increase monotonically with VPA dosing (Bai et al., 2017; Xu et al., 2019). Therefore, the paradoxical reduction in TAG_{ld} at high VPA dosing is likely due to the HEPATOKIN1 model not accounting for these physiological feedback mechanisms, particularly gene-regulatory responses, that are presumably activated under high-dose conditions.

3.2 Incorporation of PPAR α -mediated regulation is essential to predict monotonic increase in lipids

Next, to demonstrate the necessity of incorporating feedback mechanisms into our integrated model, we examined the role of the transcription factor PPAR α , given its critical functions in regulating fatty acid and lipid metabolism (Maldonado et al., 2018). We semi-quantitatively modelled the impact of PPAR α



exclusively on the metabolic enzymes it regulates and subsequently performed simulations in a single population representative virtual individual with varying VPA doses (see Methods). As expected, integrating $PPAR\alpha$ mechanism revealed no alterations in the val_{cyt} dynamics compared to the outcomes without it, as shown in Figure 3A. We also observed a dose-dependent decrease in v_{CPT1} as expected, attributed to competitive inhibition from rising val_{cyt} levels (Figure 3B). However, the inclusion of $PPAR\alpha$ resulted in higher quantitative levels of v_{CPT1} across all doses than those predicted without $PPAR\alpha$ (Figure 2B) likely from CPT1 upregulation by $PPAR\alpha$. Similarly, there was a dose-dependent increase in $c16_{cyt}$ (Figure 3C) due to reduced v_{CPT1} , yet the levels were quantitatively lower than those simulated without $PPAR\alpha$ (Figure 2C). Surprisingly, the incorporation of $PPAR\alpha$ ensured a consistent dose-dependent increase in TAG_{ld} with escalating VPA doses (Figure 3D), a behaviour absent when $PPAR\alpha$ was not included. It is important to mention that the predicted hepatic TAG_{ld} concentrations are roughly 10-fold higher than the TAG concentrations observed in patients' plasma (typically ~1.3–1.8 mM) (Sidhu et al., 2017; Aksoy D et al., 2015). The reason may stem from the larger pool of TAG in the liver compared to that present in plasma (Farquhar et al., 1965).

While the model includes packaging of TAG into very low density lipoprotein (VLDL) and hepatic VLDL secretion, it does not currently predict plasma TAG. Incorporating systemic distribution, peripheral uptake, and plasma clearance kinetics, along with refining related parameterization, would enable prediction of plasma TAG and may help reconcile the higher hepatic TAG_{ld} simulations with observed plasma measurements.

The predicted baseline for $PPAR\alpha$ remains in the range of 1.9–2.0, even in the absence of VPA (Figure 3E), likely due to diurnal variations of $c16_{cyt}$ under physiological conditions, highlighting an active basal level of gene regulation essential for maintaining homeostasis. Although the increase in $PPAR\alpha$ with rising VPA doses is modest, it is sufficient to elicit a consistent dose-dependent response in metabolite levels. Moreover, examining the cofactors of TAG_{ld} with the inclusion of $PPAR\alpha$ reveals distinct dose-dependent dynamics: $c16CoA_{cyt}$ increasing (Supplementary Figure S5C), while CoA_{cyt} decreasing correspondingly (Supplementary Figure S5D). Notably, the integration of $PPAR\alpha$ successfully addresses the complete depletion of the cofactor CoA_{cyt} , a limitation previously observed in model simulations lacking $PPAR\alpha$ regulation. This adjustment underscores the crucial role of $PPAR\alpha$ in maintaining cofactor stability across varying VPA dosages.

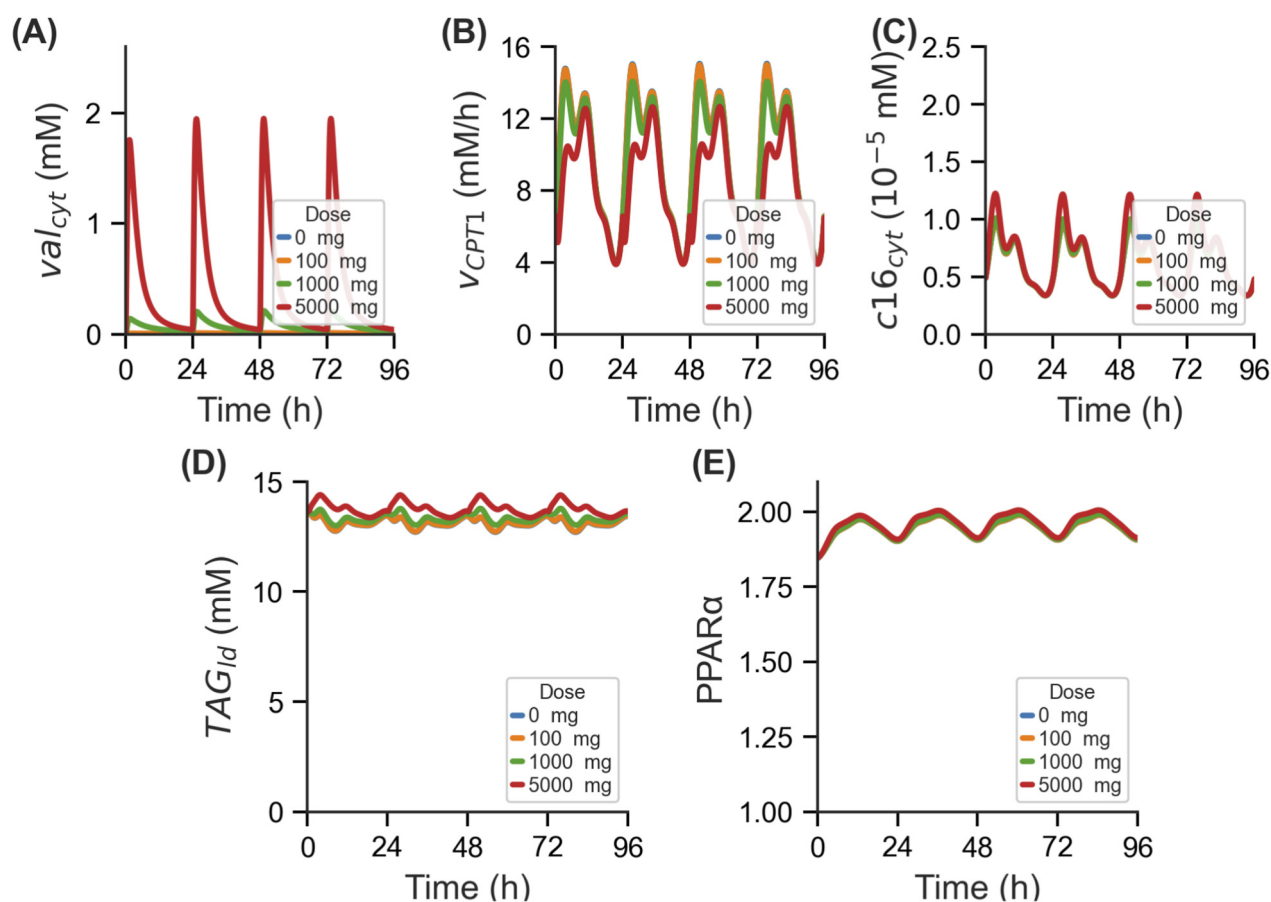


FIGURE 3

Effect of PPAR α -mediated regulation on VPA induced changes in lipid metabolism. Time-course profiles in a single population representative virtual individual for (A) free hepatocellular concentration of VPA (val_{cyt}), (B) flux of CPT1 enzyme reaction (v_{CPT1}), (C) concentration of cytosolic palmitate ($c16_{cyt}$), (D) concentration of triglyceride content (TAG_{id}), and (E) transcription factor PPAR α regulation with various repeated intravenous dosing of VPA. PPAR α , peroxisome proliferator-activated receptor α ; VPA, valproic acid.

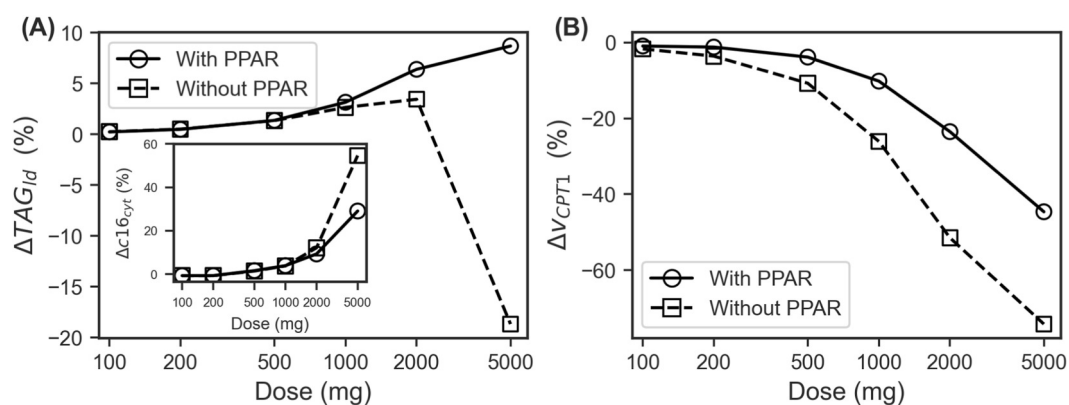


FIGURE 4

Dose-response curves of maximum % changes in (A) lipid droplet TAG content (ΔTAG_{id}) with a figure inset of the dose-response curve of % changes in maximum cytosolic palmitate concentrations ($\Delta c16_{cyt}$) and (B) maximum % changes in CPT1 enzyme velocity (Δv_{CPT1}) compared to no dose outcomes with increasing doses of once daily valproic acid. CPT1, carnitine palmitoyltransferase 1.

Expanding upon these findings, we evaluated the maximum relative change in TAG_{ld} for various VPA dosages both with and without $PPAR\alpha$, compared to a no-treatment scenario. This analysis involved determining the relative change at each time point between treated and untreated conditions for each dose and plotting the maximum predicted relative change for each dosage. The dose-dependent responses, summarized in Figure 4A, indicate that model predictions incorporating $PPAR\alpha$ retain a consistent increase in TAG_{ld} across dosages, while predictions without $PPAR\alpha$ exhibited a marked reduction in the percentage change in maximum TAG_{ld} content at 5,000 mg VPA. This model behaviour holds even as both models show a consistent dose-dependent increase in the percentage change in maximum $c16_{cyt}$ (Figure 4A inset). Furthermore, as shown in Figure 4B, both models predict a dose-dependent reduction in v_{CPT1} , owing to the competitive inhibition of VPA. However, the reduction in v_{CPT1} is noticeably less pronounced in the model with $PPAR\alpha$ at doses 1,000 mg and above compared to those without. This difference can be explained by the upregulation of CPT1 by $PPAR\alpha$, suggesting $PPAR\alpha$ mitigates the reduction in v_{CPT1} resulting from VPA inhibition.

4 Discussion

This study aimed to apply and extend the hepatocyte metabolism model for assessing early markers of liver steatosis, focusing on CPT1 inhibition by VPA and evaluating dose-dependent changes in various hepatic metabolites. We integrated three different modelling components: PBPK, a large-scale hepatocyte metabolism model (HEPATOKIN1), and $PPAR\alpha$ -mediated regulation, forming a tool-based and multiscale QST model. PBPK integration enabled the simulation of physiologically relevant VPA concentrations, driving downstream effects within the hepatocyte metabolism model. Our findings demonstrated that competitive CPT1 inhibition of VPA alone could induce dose-dependent increases in lipids, including TAG, as clinically observed before (Nanau and Neuman, 2013). However, the addition of $PPAR\alpha$ regulation into the hepatocyte metabolism model was essential to address the paradoxical reduction of liver metabolites at high VPA doses, underscoring the importance of considering known biological regulations likely to be activated under extreme scenarios such as high drug dosing.

An increase in circulating TAG has been observed in epileptic patients undergoing VPA treatment, with reported increases ranging from approximately 10%–50% of circulating TAG (Sidhu et al., 2017; Aksoy D et al., 2015). In contrast, our model predicts TAG within hepatocyte lipid droplets, with a maximum TAG increase of about 10% (Figure 4A). This discrepancy may arise from several factors, including comparing circulating TAG against TAG within hepatocyte lipid droplets, patient demographics (such as age (Guo et al., 2022) and BMI), treatment duration, and the presence of insulin resistance (Ozdemir et al., 2011; Pylvanen et al., 2003). The reported increases in TAG due to VPA treatment typically manifest after 1–2 years, whereas our simulations only spanned a few days. Prolonged treatment durations may have unforeseen implications on additional metabolic pathways or regulatory networks. Furthermore, many patients receiving VPA monotherapy develop insulin resistance, which can significantly impact metabolic pathways (Sidhu et al., 2017; Pylvanen et al., 2003). Our model does not

account for the influence of insulin on TAG synthesis, which may exacerbate effects within the metabolic network and explain the higher observed changes than predicted in our simulations.

There are additional limitations associated with the current implementation of the integrated model that must be recognized. The exploration of $PPAR\alpha$ -driven regulation in this study is not comprehensive, due to the complex and wide-spread influence of $PPAR\alpha$ (Kersten, 2014), as well as uncertainties surrounding its effects on specific enzyme kinetics. Additionally, there is a paucity of quantitative data on $PPAR\alpha$ effects, and human-specific data is even more limited (Kersten and Stienstra, 2017), which creates a challenge in comprehensively modelling the regulatory effects of $PPAR\alpha$ on individual enzymes. An important interspecies limitation is that $PPAR\alpha$ activation in rodents is generally higher than in humans, with more pronounced transcriptional responses and metabolic modelling reported in rodents (McMullen et al., 2020). Consequently, extrapolating $PPAR\alpha$ -mediated any insights from rodent studies to clinical observations warrants caution in interpretation. Nevertheless, our model provides a framework that can be re-parameterized with human-specific data to support cross-species translation, and the integration of a human VPA PBPK model validated against clinical data enables exploration of human-driven PK effects as a first step toward such translation. Similar challenges apply to the hepatocyte metabolism model, which relies on chimeric data comprising inputs from various preclinical studies and species to inform or optimise parameters. Consequently, our focus was on the qualitative direction of effects when interpreting the impact of VPA, rather than precise quantitative predictions.

In this study, we focus on $PPAR\alpha$ -driven regulatory effects, given the extensive research linking $PPAR\alpha$ specifically to liver pathways (Jump et al., 2013). Consideration of other regulatory pathways fell outside the scope of this study. However, it is important to acknowledge that numerous other regulatory pathways can influence metabolic pathways in the hepatocyte metabolism model. These include other $PPAR$ subtypes ($PPAR\beta/\delta$, $PPAR\gamma$) (Kersten and Stienstra, 2017), other transcription factors (Jump et al., 2013; Han et al., 2016; He et al., 2025) such as HNF4, retinoid X-receptor, liver X receptor, NRF2, and proteins (Yamashita et al., 2001) like ChRE-binding protein. Therefore, when utilizing this QST model or similar models, it is essential to evaluate the relevance of additional regulatory pathways pertinent to the specific research hypothesis and adapt the model accordingly.

The limitations discussed above can be addressed through further quantitative measurements in specific metabolic pathways, which would enhance the translatability and clinical relevance of the QST model. For instance, advancing our understanding of liver metabolism pathways specific to humans, with quantitative outcomes, can provide direct insights for informing parameters or optimizing them to replace those derived from preclinical species. This refinement would significantly improve the model's applicability in clinical settings. Moreover, incorporating additional regulatory pathways involves identifying the relevant liver metabolism pathways affected, determining the direction of regulation (up or down), and establishing connections to relevant metabolites, similar to the inclusion of $PPAR\alpha$ included in this work. To further increase confidence and applicability, integration of clinically relevant, non-invasive DILI biomarkers will be important to enable prospective validation of model predictions. In parallel, expanding the set of test

compounds with available clinical data will help reduce parameter uncertainty, improve identifiability, and strengthen the robustness of quantitative estimates. Therefore, as our understanding of VPA effects and hepatocyte metabolism, along with regulatory effects, continues to expand, the model demonstrated here should be iteratively updated to incorporate relevant mechanisms essential for exploring specific research hypotheses.

This work demonstrates the potential of multi-scale *in silico* modelling as part of the new approach methodologies (NAMs) landscape. By integrating PBPK, a large-scale hepatocyte metabolism model, and PPAR α -mediated regulation into a QST model, this study demonstrates a mechanistic, human-relevant approach to predicting adverse outcomes. This computational framework offers a powerful alternative to traditional animal testing, focusing on understanding the underlying biological mechanisms of toxicity (e.g., CPT1 inhibition) through the integration of multi-modal data and striving for greater translatability to human health. Such integrated modelling efforts are crucial for enhancing early hazard identification, refining risk assessment, and ultimately reducing reliance on animal models in toxicology and drug development.

Data availability statement

Simcyp Designer and Simcyp Simulator model workspaces to generate VPA effects on hepatic metabolism, Simcyp Simulator workspaces for VPA validation, and their curated Excel outputs are available on the platform BioModels (Malik-Sheriff et al., 2020) as MODEL2509090001. Simcyp Designer and Simcyp Simulator software can be licensed from Certara Predictive Technologies, and a license to reproduce outcomes can be provided upon request.

Author contributions

VG: Conceptualization, Investigation, Writing – original draft, Writing – review and editing. HJ: Methodology, Writing – original draft, Writing – review and editing. CF: Conceptualization, Investigation, Writing – original draft, Writing – review and editing. AK: Conceptualization, Writing – original draft, Writing – review and editing.

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Conflict of interest

VG is currently employed at AstraZeneca and conducted this research as an employee of Certara Predictive Technologies and the views expressed do not necessarily represent the views of AstraZeneca. VG may hold stock in both companies. CF, currently employed at GSK, contributed to this article as an employee of Certara Predictive Technologies and the views expressed do not necessarily represent the views of GSK. CF may hold stock in both companies. AK and HJ are current employees of Certara Predictive Technologies and may hold stock in the company.

Generative AI statement

The author(s) declared that generative AI was used in the creation of this manuscript. Finally, we acknowledge the use of OpenAI's ChatGPT-4.1 for language polishing during the writing process.

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Supplementary material

The Supplementary Material for this article can be found online at: <https://www.frontiersin.org/articles/10.3389/fphar.2026.1768190/full#supplementary-material>

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